Lab 05 - Op Amp Circuits I (DAC)

Objective

This lab is to familiarize students with the operations and usage of op-amps and the fundamentals of a digital-to-analog converter

Introduction

In this lab you will build an op-amp circuit to convert digital numbers to analog voltages. This circuit is called a digital-to-analog converter, or DAC.
Numbers can be represented electronically in digital form or analog form. However, for most of our work, we prefer to use the binary representation of numbers since we can handle these with digital logic circuits, in which a 1 corresponds to a high voltage, and a 0 corresponds to a low voltage (or vice-versa, if we choose). As an example, the number “100” can be represented in binary as: 0110 0100. This corresponds to the decimal value via the following:

$0*2^7=0 \ + \ 1*2^6=64 \ + \ 1*2^5=32 \ + \ 0*2^4=0 \ +$
$0*2^3=0 \ + \ 1*2^2= 4 \ + \ 0*2^1= 0 \ + \ 0*2^0=0 \ = 100$

Discrete versus Continuous

Digital representation produces discrete values. The word "discrete" means there are only certain allowed values that can be represented. For example, 2 binary bits can only handle 4 discrete values (0-3) while 3 binary bits can handle 8 discrete values (0-7). Thus, the more bits, the higher the resolution. The word resolution is commonly used in two ways- in one usage it means the number of bits of encoding, but it is also used to mean the voltage difference between discrete steps. For example, for 3-bit encoding, we could say our system has "3-bit resolution," or we could say the "resolution is 0.125 Volts" (assuming 1V full-scale). The latter usage is also called the least-significant bit, or LSB. For example, for the 2-bit encoding, we could say "the LSB corresponds to 0.25 Volts" (1V full-scale).
Note, since the discrete values are 0-based, the V_{max} represented in binary never reaches the full-scale value since the maximum is always one LSB below full-scale. We can also see that the resolution (or LSB) is determined by the number of bits represented by:

$$R = LSb = V_{FS}\left(2^{-N}\right)$$

where R is the resolution in volts, N is the #bits, and V_{FS} is the full-scale voltage.
Analog representation is continuous, which simply means that all values are allowed. However, in practical circuits the resolution is not zero. Instead, it is the smallest change of voltage that can be reliably detected, which depends on the amount of electronic noise that is present and on the precision of the various components.
Early computers used analog representation, and performed all calculations without converting to digital. The computers were called analog computers, and the circuits they used for performing arithmetic operations were called operational amplifiers, or op-amps. These same op-amp circuits were refined over the years and are the ones we now use for many applications such as amplification, buffering, filtering, and digital-to-analog conversion.

Digital to Analog Converters (DACs)

Digital-to-analog converters (DACs) are circuits that have an N-bit digital value as input, and generate the corresponding analog voltage (or current) as output. They are most often used to convert computer-generated data into real-world signals such as sound and images.
Analog–to-digital converters (ADCs) perform the opposite function - they convert an analog input voltage to a digital format. ADCs are most often used to convert real-world signals into binary form for processing or storage by a computer system.
In this experiment we will build and test a 4-bit DAC.

Theory: R-2R Ladder DAC

The DAC we will build in this experiment is based on an op-amp circuit called the R-2R ladder. This is an extremely clever circuit invented during the 1950's. It was considered unique enough to earn a patent! A 4-bit R-2R circuit is shown below (Prelab section). The resistance R can be any value, but it is usually chosen to be large to reduce the current required from the V_{FS} source.
In this circuit, the binary digital input is provided by the switches S_1 through S_4. Each input terminal is connected to either V_{FS} (logic 1) or to ground (logic 0). The most significant bit (MSB) is input B_4 provided by switch S_4. When all the input bits are 0, V_{OUT} = 0. By direct circuit analysis, one can calculate the change in output voltage contributed by each input. The results for our 4-bit circuit are shown in Table 1.

Bit	**Output voltage change, ΔV_{OUT}**
B_4 (MSB)	$-\frac{1}{2}V_{FS}$
B_3	$-\frac{1}{4}V_{FS}$
B_2	$-\frac{1}{8}V_{FS}$
B_1 (LSB)	$-\frac{1}{16}V_{FS}$

Table 1. Change in V_{out} when each input is switched from logic 0 (GND) to logic 1 (V_{FS})

The output voltage change is negative, because the op-amp is used in the inverting configuration. Using the principle of superposition, we can derive the analog output for any combination of input bits. The result is expressed as:

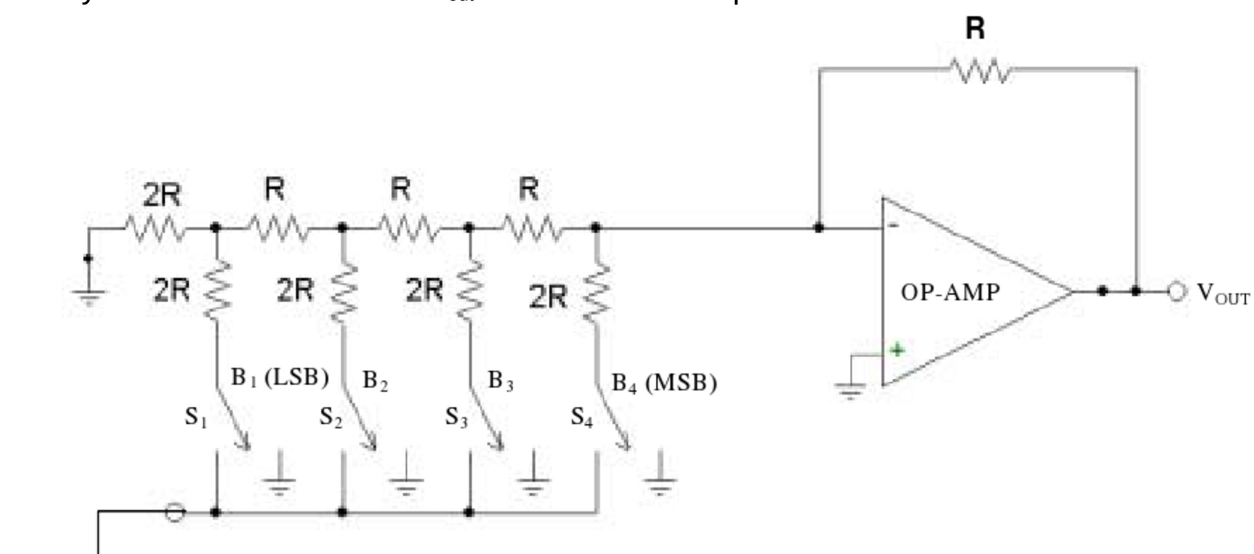
$$V_{out} = -(\text{Binary Value})(2^{-N})V_{FS}$$

where "Binary Value" is the number represented by $B_4B_3B_2B_1$**,** and N is the number of bits.

Prelab

Design an LTSpice simulation that implements a 4-bit DAC similar to the circuit shown below:

- Use the "ideal" op-amp (component "opamp") which only has the 3 terminals: Inverting input, Non-inverting input, and Output.
- Technically, R values can be anything, but typically chosen to be large in order to reduce the current required from the V_{FS} source. Run the following simulations:
  - Choose R=50kΩ & 2R=100kΩ to get the proper R/2R ratio
  - Choose R=47kΩ & 2R=100kΩ to get a modified R/2R ratio using accessible parts
- You will need to include the SPICE directive ".include opamp.sub" on your design
- Verify the simulations create V_{out} that follows the equation above



The position of each switch is determined by the input bits as follows:

- If bit $B_i = 1$, then switch S_i is connected to V_{FS}.
- If $B_i = 0$, then S_i is connected to ground.

In-lab Procedure

Parts needed:

- Four 47kΩ resistors
- Five 100kΩ resistors
- One 741 (or equivalent) op-amp
- A selection of colored 22 gauge connection wires

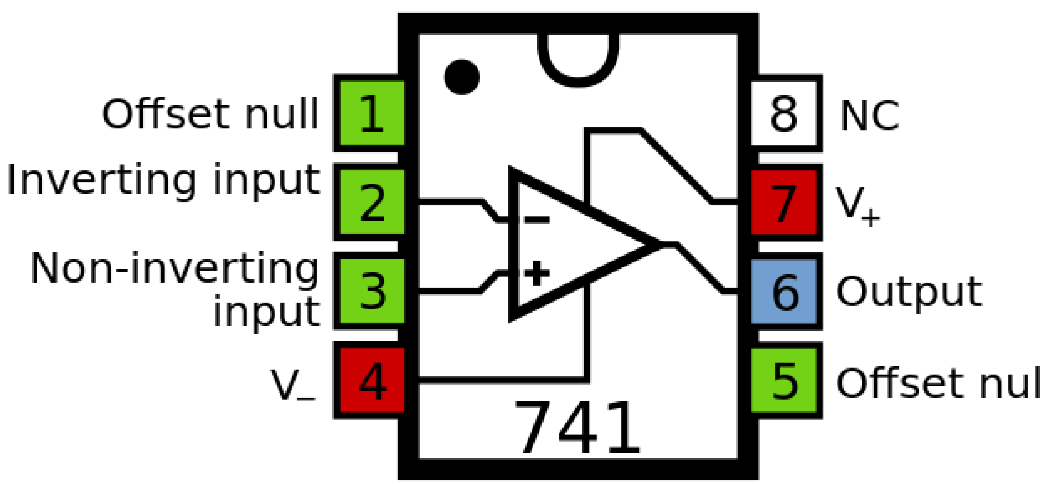
It is a good idea to always verify your component values before you begin doing an experiment. This will allow you to identify one possible source of error easily.

Build and Measurements

- Plug in your XK700 trainer but **leave the power off**
- Build the DAC on the breadboard in your XK700 (see 741 pinout below)
- For the switches, use the output terminals that are connected to the SW1-SW4 data switches on the XK700: down=LOW (0V) and up=HIGH (5V)
- Use GND from the Power Supply (PS) for your ground points
- Use the +12V/-12V from the PS output terminals to connect to V+/V- on the 741 op-amp.
- **Double-check your connections before powering up the system!**
- Measure and record the output voltage for all digital values between 0000b and 1111b.

Lab Report Requirements

- Submit a lab report based on the lab report template provided on Blackboard.
- Potential Discussion Items
  - How did changing the R resistor to something less than 1/2 of 2R change the results? Is there a way to compensate for it by using the feedback resistor? In your lab report, discuss some ideas on a modification to the R-2R ladder design that utilizes actual resistor values available to you that would still produce equivalent results to the theoretical R-2R ladder (within 0.5%).
  - Compare your actual results to the ideal R/2R simulation. How much was the "error"? What modification(s) would you propose to the design in order to get the actual circuit to approach the theoretical, given the resistors available to you?
  - Compare your actual results to the modified R/2R simulation. Did the SPICE model accurately predict the voltages? What was the error? Discuss any discrepancies.



*Generic **741 Pinout***